Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Reference electrodes, calomel electrode

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Reference electrodes, calomel electrode
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Reference electrodes, calomel electrode

1 Reference electrodes, calomel electrode

Study focus: Reference electrodes, calomel electrode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module 4 on electrochemical equilibrium.

Key details

- And in today's class we are going to look at reference electrodes.
- That would be the primary reference electrode which is your standard hydrogen electrode and then we will study about the secondary reference electrodes which are galomyl electrode and silver chloride electrode.
- We know from the previous classes that whenever we form a cell, it will be two electrodes which are coupled together.
- What we can measure is just the potential difference.
- and measuring the electrode cell potential we will be able to determine the potential of the other electrode.
- This other electrode is known as the indicator electrode.
- Then we combine the reference electrode with the indicator electrode.
- You're familiar with this, they are called the primary reference electrode and the secondary reference electrode.
- And for secondary reference electrodes we have.
- Calamile electrodes and the elect and the silver chloride electrode.
- As I mentioned just now, the standard hydrogen electrode, which is also referred to as the SHE, is used as a primary reference electrode.
- It is assigned a value of 0.0 volts.
- It is by convention and internationally we accept that electrode potential for a standard hydrogen electrode is assigned a value.
- of 0.0 volts.

- We have a gas being bubbled around an inert electrode which is dipped in a solution of ions to which the gas is reversible.
- as we studied yesterday.
- Always one atmosphere and the H plus concentration is maintained at one molar concentration.
- As mentioned earlier, we have to form a cell.
- Zinc rod dipped in Z2 plus ions.
- How do I know this is a standard hydrogen electrode?
- Because I've used one atmosphere pressure and one molar concentration of H plus ions.
- And we know that E cell is always E right hand side minus the E left hand side, which is E cathode minus E anode.

Example or application from the lecture

- Let us first define reference electrodes.
- An example for primary reference electrode in fact this is the only electrode which is used as a primary reference electrode is the standard hydrogen electrode.
- So let us look at first what is the primary reference electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

And this is equal to 0, as I said mentioned just now that standard hydrogen electrode potential is always given a value of 0 minus E of the zinc electrode.

Key details

- Since it is a positive value, I know that the connections I have made here are right, meaning what that zinc electrode will act as the anode and the hydrogen electrode is going to act as a cathode.
- If there was any mistake in this, if I had put hydrogen electrode here and the zinc electrode here then I would have measured E cell as a negative value but since I got a positive value I know that zinc will act as anode here and hydrogen electrode will act as the cathode.
- Zn Zn2 plus.
- Are also at one molar concentration.
- Nevertheless, from this arrangement, it can be any concentration of zinc, doesn't matter.
- I can get the value of the electrode potential if I can measure the cell potential.
- Any electrode you want to measure the potential, whatever cell potential you measure is the value which is to be assigned to that particular electrode.
- Why do we need any other type of reference electrode?
- Continuously to maintain it at that particular value is very difficult.
- So that is the first problem we face, and then we have platinum.
- Which is highly susceptible to poisoning.
- We know that platinum gets poisoned easily by the different impurities which could be present in the gas, and the third point is that the high standard hydrogen electrode cannot

be used wherever there is an oxidizing and reducing energy.
- Therefore, these are the problems associated or disadvantages or limitations of the standard hydrogen electrode.
- Hence we have to look for other type of electrodes which will act as reference electrode.
- Now for a secondary reference electrode because we have these limitations with the standard hydrogen electrode, all these problems should be removed now.
- Apart from that.
- Of course, this is very important, the electrode potential should be accurately known and it should remain stable.
- for a long period of time.
- Over a long period of time we should have a stable potential so that when we measure the potential difference we know that this electrode has this potential and therefore the other part the difference of this.

Example or application from the lecture

- The reason is that the standard hydrogen electrode has certain disadvantages so what are the disadvantages let us have a look at it now maintaining H plus ions at one molar concentration and the pressure of hydrogen at one atmosphere is very difficult so.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

potential difference from the electrode potential of the reference electrode would give me the electrode potential of the indicator electrode so one thing it should be accurately known second it should be stable over a long period of time and it can be easily assembled so these are the three things which we look for when we design or select a secondary reference electrode.

Key details

- And with this, of course, since it is a reference electrode, my definition gets complete when I say electrode potential of other electrodes can be assigned with the help of this electrode.
- Come under this category one is the caramel electrode, the other is the silver chloride electrode.
- Now the first thing if you are asked to describe the Calomel electrode and explain the Calomar electrode first thing we need to write or we need to say is that it is used as a secondary reference electrode and it is a very widely used secondary reference electrode.
- And then the next thing, what category does this electrode fall in?
- Metal, insoluble salt.
- Because in a calomyl electrode I know.
- I have to use a metal that is mercury is used, then I need to use H2Cl2, which is the insoluble salt or sparingly soluble salt of the metal, and then I need to use the KCl solution.
- And we place a layer of mercury in it.
- Above this layer.
- We take a paste of mercury and HG2Cl2.

- Hg and over that now we place KCl solution.
- You can see a platinum wire is dipped here and it will go into the mercury.
- Platinum wire, it is providing the electrical contact.
- Mercury being a liquid, we don't have an electric contact electrical contact established here.
- outlets or inlets over here.
- This inlet is for introducing KCl solution as in when we need to replace it and this is connecting to the salt bridge.
- This tube over here outlet for the solvent this is connecting to the solder so this is the construction or this is what your calomel electrode looks like.
- Hg, H2, Cl2 solid that is and Cl minus.
- Because that is true for any electrode, it always depends on the other electrode which is being combined.
- It reacts with the chloride ions which we have provided there in the form of KCl and this gives you HE2Cl2.
- What is it?
- We add this to and what we get 2 H G plus 2 Cl minus is equal 2 Cl minus is equal to H E two Cl2 plus 2 electrons.

Example or application from the lecture

- So we see that mercury first forms a G22 giving out electron because it's an anode so electrons are being given out over here.
- This is the reaction when calomel acts as anode, and you can see electrons are being given out here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

is giving you 2 Hg and then H2Cl2 undergoes dissociation to compensate for the G2 plus which have been used up here.

Key details

- When the caromel electrode acts as cathode, again it is true we can understand because two electrons here, the electrons are being taken up, so it's a reduction reaction.
- Writing the Nernst equation.
- For the Nernst equation, what do we need?
- We need the Q value first.
- We can see H G raised to the power of 2, Cl minus raised to the power of 2 divided by HE2 CL2.
- Are pure form, so the value will be one for them.
- Therefore, the q value is just Cl minus squared.
- Okay, so one now we write the Nest equation E is equal to E naught minus two point three zero three R T over two f two is because of these two electrons over here log of Cn minus square.

- We get the expression that is E is equal to E naught minus 0.0591 log of Cl minus.
- When we get this expression, what we say is the electrode is reversible to chloride ions.
- What does it mean?
- That the electrode potential.
- Is depending on the chloride ions, or in other words, if we know the electrode potential of this electrode, we will be able to determine the chloride ion concentration.
- It's very obvious from this equation E is equal to E naught minus 0.059 1 log of Cl minus so electrode potential depends on the chloride ions.
- But we did not specify what is the concentration of KCl we are using.
- So since now we understand that the electrode potential is going to depend on the chloride ion concentration, that means the KCl which is there in the electrode.
- The potential is going to depend on the concentration of the KCL which we have used.
- If we use 0.1 molar KCl or 0.1 normal, then it will be known as a decinormal electrode and the potential is 0.3358 volts.
- And if we use a one molar concentration on one normal, then it is a called a normal electrode and it is 0.2824 volts, and whereas if it is a saturated solution of KCl and then it is known as a saturated calom electrode and the electrode potential is 2244 volts.
- Silver chloride electrode again is widely used as a reference electrode.
- So that's the first thing we need to note.
- Because mercury being a liquid, we took a tube and we placed mercury.
- And then in that we place a platinum wire for electrical contact.
- But here we don't need to do that because silver is already in the solid form.
- Or we can have a silver clore coated platinum wire.
- It is a metal insoluble salt ion electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 20:00–24:59

AGCl that is silver chloride which is electrolytically coated over the silver wire and now we need KCl solution.

Key details

- H Cl is also allowed, we can use HCl also of known concentration.
- So Ag plus electron is giving you AG and AG Cl gives you Ag plus Cl minus and therefore overall reaction would be A G Cl plus electron is giving you AG plus Cl minus.
- If you remember when I taught second type of electrodes that is metal insoluble salt.
- Ion electrode I gave this reaction over there and also mention it is not AG plus it is AG and it is balanced with the negative charge is balanced in this way so now we know how we got this reaction so it is ag Cl plus electron is giving you.
- A G plus the Cl minus.
- N is 1.
- potential is going to depend on the chloride ion concentration.

- Based on that we have three different types of silver chloride electrode because again why the chloride ion concentration is coming from KCl or if you have used HCl then HCl so based on the chloride ion concentration if it is 0.1 normal Kl.
- A Cl which you have used, then it is known as a decnormal electrode and the value is electrode potential at 298 Kelvin is 0.289 volts.
- If you use one normal, then it is known as a normal electrode and the electrode potential is 0.223 volts.
- And if you use a saturated solution of Caseyl then it will it will be referred to as it will be called saturated silver chloride electrode.
- And the potential is going to be 0.199 volts.
- Silver chloride electrode also depends on the chloride ion concentration which we have used.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Reference electrodes, calomel electrode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reference electrodes, calomel electrode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

C Revision and retrieval practice